

Stone Speaks: An Embodied AR-IoT System for Revitalizing Immovable Heritage

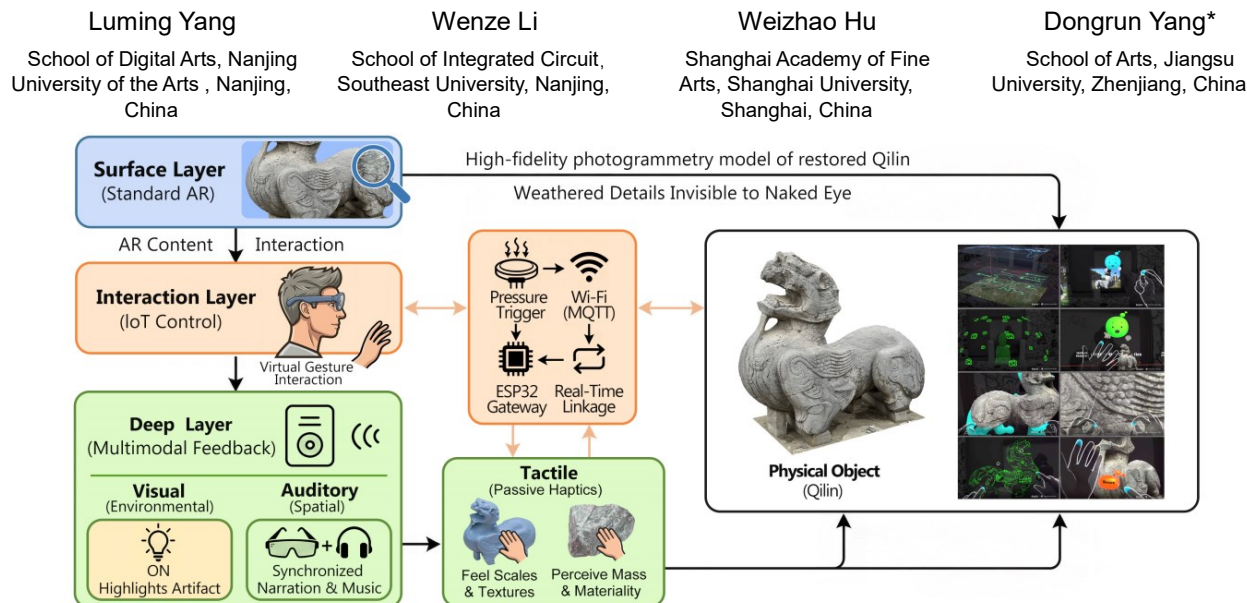


Figure 1: System architecture of the Closed-Loop AR-IoT Interaction framework. Pressure-based interactions trigger an ESP32 gateway via the MQTT to deliver synchronized multimodal feedback (lighting, spatial audio, and passive haptics), creating a closed-loop immersive experience.

ABSTRACT

Immovable cultural heritage faces exhibition challenges due to spatial constraints and limited interactivity. We present a Closed-Loop Augmented Reality and Internet of Things (AR-IoT) Interaction model combining Augmented Reality (AR) visualization with IoT environmental control. Using Southern Dynasty stone carvings as a case study, our system employs AR glasses with pressure-switch-triggered interactions to control on-site lighting and spatial audio, achieving a closed-loop virtual-physical experience. An evaluation with 25 participants demonstrated that AR+IoT integration significantly enhances the sense of presence ($t(24)=8.3, p<0.001$) and cultural comprehension ($t(24)=5.2, p<0.001$) compared to Standard AR, validating the effectiveness of coordinated multimodal feedback for heritage interpretation.

Index terms: Cultural Heritage, Augmented Reality, Internet of Things, Embodied Interaction, Mixed Reality.

1 INTRODUCTION

While Augmented Reality (AR) and mixed reality technologies

have demonstrated potential for enhancing museum visitor experiences [1, 2], existing applications often lack embodied engagement. Most solutions remain restricted to unidirectional visual overlays, failing to activate the multi-sensory channels required for deep immersion [3]. These systems do not establish dynamic dialogue between visitors and physical spaces, limiting their effectiveness for cultural heritage interpretation. Building on embodied cognition theory [3] and the reality-virtuality continuum framework [4], we propose a closed-loop AR-IoT interaction model applied to the monumental Southern Dynasty stone carvings (420–589 CE) in China, widely scattered field relics that currently lack effective in-situ interpretation mechanisms to engage visitors. By integrating AR visualization with pressure-switch-triggered environmental control (lighting and spatial audio), we create a "Reality-Virtual-Reality" feedback cycle that significantly enhances immersion and cultural understanding. The core architecture of this system builds upon the framework described in prior work [5]. This study presents the system implementation and reports new quantitative evaluation results regarding user presence and comprehension. Our contributions include: (1) a novel Closed-Loop AR-IoT Interaction architecture for immovable heritage sites, (2) a closed-loop interaction design combining visual, auditory, and tactile modalities, and (3) empirical evidence demonstrating enhanced presence and comprehension compared to Standard AR approaches.

2 SYSTEM DESIGN

2.1 Theoretical Framework

We operationalize embodied cognition by directly linking physical actions, such as tactile exploration, to environmental feedback, rather than relying on passive visual overlays. This approach establishes a tangible connection along the reality-virtuality

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continuum, in which digital information is not merely superimposed but causally activated by the user's bodily interactions with physical replicas. Consequently, comprehension is rooted in sensorimotor experience.

2.2 System Architecture

The system operates as a closed-loop with three integrated layers (Figure 1):

Surface Layer (Standard AR): We employ the Unity engine with XR SDK to create high-fidelity photogrammetry-based 3D models of the stone carvings. Rokid AR Studio glasses with spatial anchors enable precise alignment between digital overlays and physical artifacts. A Virtual Pedagogical Agent provides voice-over explanations and directs attention to specific details, reducing cognitive load and bridging the temporal gap between ancient artifacts and modern visitors.

Interaction Layer (IoT Control): A pressure-sensitive switch mechanism ensures intentional engagement. When users actively pick up specific exhibit replicas to interact, the embedded pressure sensor triggers environmental changes. Commands are transmitted via Wi-Fi using the MQTT protocol to an ESP32-based gateway, ensuring low-latency response.

Deep Layer (Multimodal Feedback): When triggered, the IoT gateway orchestrates synchronized feedback across three modalities: (1) Visual—LED lights illuminate the physical artifact or background, (2) Auditory—spatial audio via AR glasses' built-in directional speakers delivers historical narration and period-appropriate music, and (3) Tactile—high-precision 3D-printed replicas positioned beside artifacts enable physical engagement [5, 6]. This completes the feedback cycle, grounding abstract historical knowledge in physical reality.

2.3 Interaction Design

The closed-loop interaction follows a deliberate sequence designed to progressively deepen engagement:

Phase 1 Initial Exploration: Upon approaching the physical artifact, users see AR overlays revealing weathered details of the stone carving invisible to the naked eye. The Virtual Pedagogical Agent introduces the artifact's historical context, preparing users for deeper interaction.

Phase 2 Intentional Activation: Users actively pick up the exhibit replica when ready to engage further, triggering the embedded pressure sensor. This physical act of picking up ensures meaningful interaction rather than passive observation.

Phase 3 Multimodal Response: The system executes coordinated feedback: environmental lighting highlights physical textures, spatial audio immerses users in historical narratives, and tactile exploration of replicas provides haptic grounding. This synchronized response creates a "Reality-Virtual-Reality" cycle in which physical actions trigger virtual content, which in turn triggers physical environmental changes. e (environmental lighting) the cultural heritage, engaging multiple sensory channels for deeper comprehension.

3 EVALUATION

We conducted a within-subjects study (N=25) comparing Standard AR versus AR+IoT Integration in counterbalanced order. Assessment employed an adapted Witmer & Singer [7] scale for Presence and custom knowledge checks for Cultural Comprehension. The AR+IoT condition yielded significantly higher Sense of Presence (M=6.2 vs. M=4.3; $t(24)=8.3$, $p<0.001$) and Cultural Comprehension (M=5.8 vs. M=4.5; $t(24)=5.2$, $p<0.001$), as shown in Figure 2. Post-experiment interviews validated the closed-loop design's value, with participants noting that "the lighting turning on made the sculpture feel alive" (P7) and

"the whole room responded to me" (P14), confirming that environmental feedback is crucial for embodied heritage interpretation.

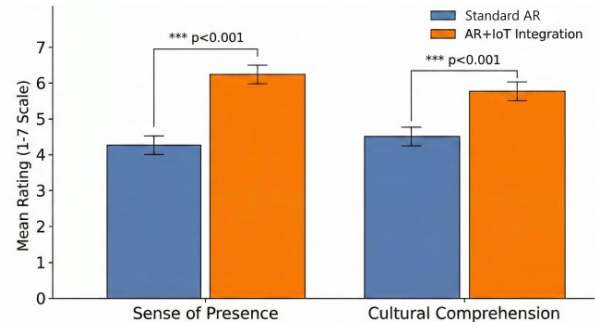


Figure 2: Comparative evaluation results (N=25). The AR+IoT condition (Orange) shows significantly higher ratings in Sense of Presence and Cultural Comprehension compared to the Visual-only baseline (Blue). Error bars represent standard deviation (***) $p < 0.001$.

4 CONCLUSION

This work demonstrates that the Closed-Loop AR-IoT Interaction approach effectively addresses exhibition challenges of immovable cultural heritage. By orchestrating synchronized visual, auditory, and tactile feedback loops, our system transcends the limitations of "screen-only" AR. Evaluation results confirm significant improvements in presence and comprehension, validating the efficacy of closed-loop multimodal interaction for heritage interpretation. Key advantages include non-invasive deployment and modular scalability. Future work will explore outdoor sites using solar-powered IoT nodes, inclusive design for visually impaired visitors, and longitudinal studies to assess learning outcomes.

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